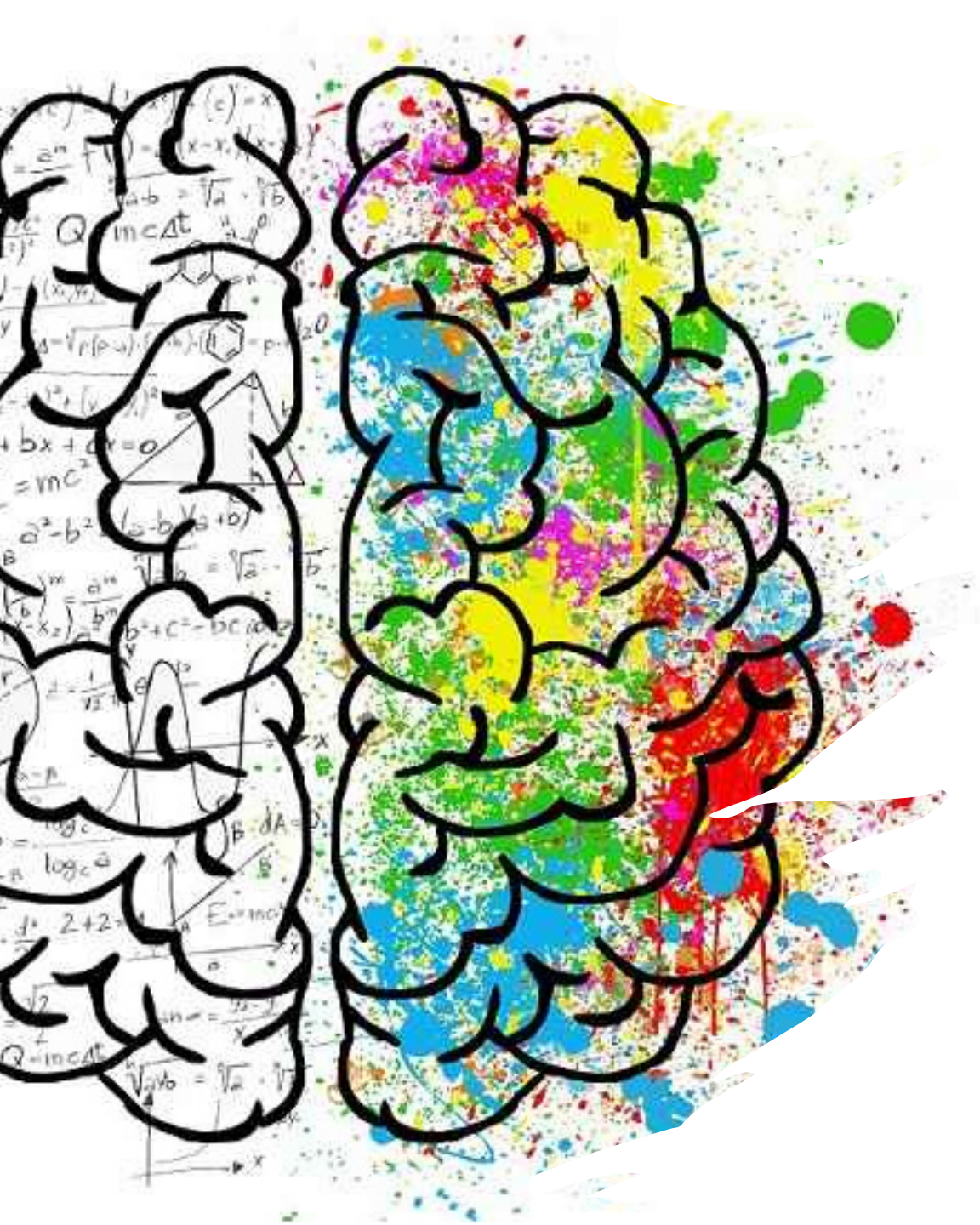




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THINKING AND ANALYSIS



Chapter outlines

THINKING AND ANALYSIS

1. Patterns of thought
2. Creative Thinking Skills

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Identify different patterns of thought, such as those found in Bloom's taxonomy
- Discuss the relationship of each thought pattern to education
- Define creative thinking
- Identify the value of creative thinking in education
- Describe the impact of limitations (such as rules) on creative thinking

1. Patterns of thought





What Is Thought?

“We exist, and we are aware that we exist, because we think. Without thought or the ability to think, we don’t exist.”

What Is Thought?

- **Thinking** is the mental process you use to form associations and models of the world. When you think, you manipulate information to form concepts, to engage in problem-solving, to reason, and to make decisions.
- **Thought** can be described as the act of thinking that produces thoughts, which arise as ideas, images, sounds, or even emotions.

“Cogito ergo sum.” = “I think, therefore I am.”

(philosopher René Descartes, French, the early 1600s)



What Are Learning Objectives?

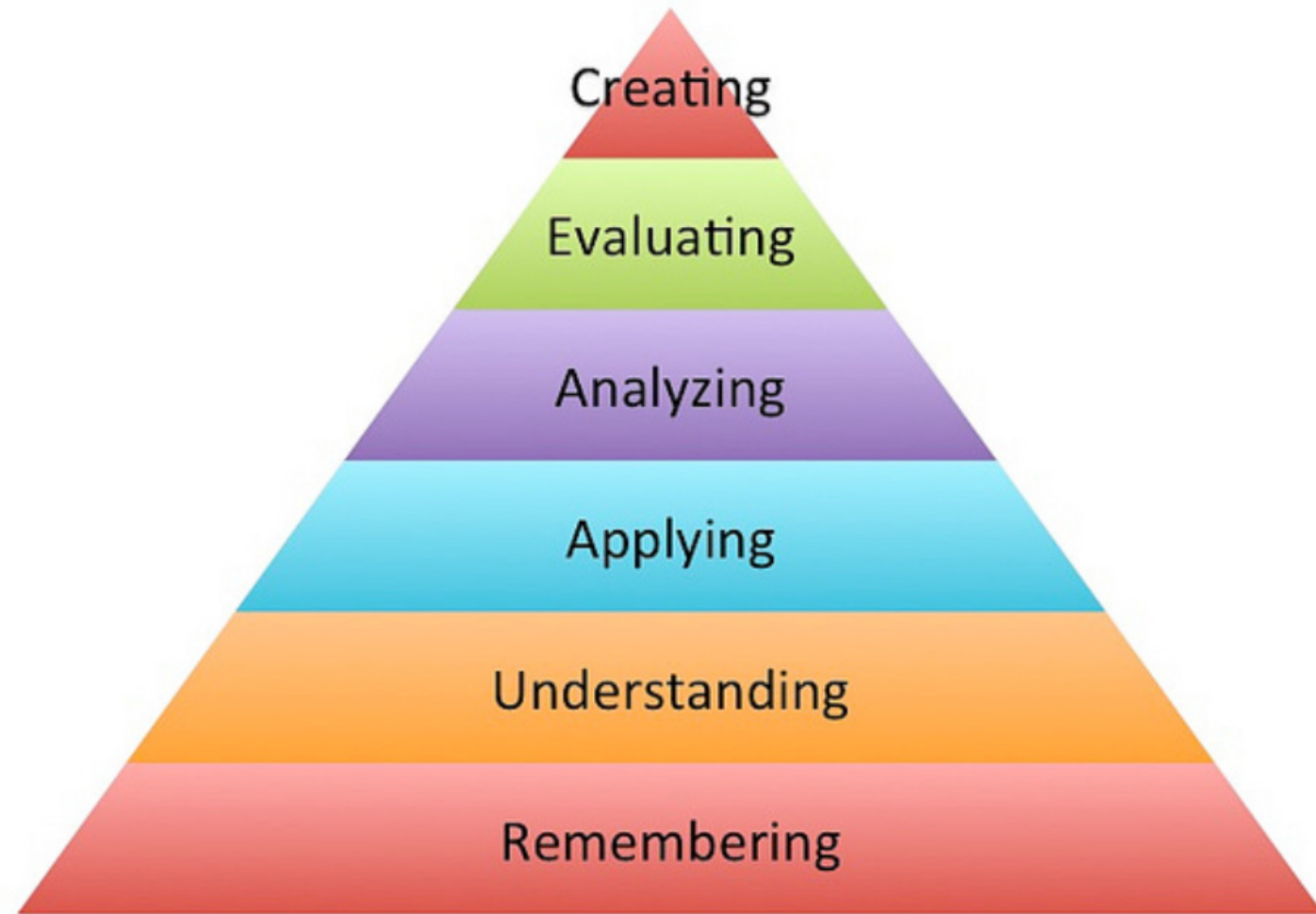
Learning objectives are *goals* that specify what someone will know, care about, or be able to do as a result of a learning experience.

The learning skills can be divided into three main categories or “domains”:

- **Cognitive domain** (*what you should know*),
- **Affective domain** (*what you should care about*)
- **Psychomotor domain** (*what you should be able to do*)



The
Cognitive
Domain of
Learning



The New Version of Bloom's Taxonomy

DESCRIPTIONS OF THE BLOOM'S TAXONOMY - THE COGNITIVE DOMAIN

MAIN SKILL LEVELS	DESCRIPTION
Remembering	When you are skilled in remembering, you can recognize or recall knowledge you've already gained, and you can use it to produce or retrieve or recite definitions, facts, and lists.
Understanding	Understanding is the ability to grasp or construct meaning from oral, written, and graphic messages.
Applying	When you apply, you use learned material (or you implement the material) in new and concrete situations.

DESCRIPTIONS OF THE BLOOM'S TAXONOMY - THE COGNITIVE DOMAIN

MAIN SKILL LEVELS	DESCRIPTION
Analyzing	When you analyze, you have the ability to break down or distinguish the parts of material into its components, so that its organizational structure may be better understood.
Evaluating	With skills in evaluating, you are able to judge, check, and even critique the value of material for a given purpose.
Creating	With skills in creating, you are able to put parts together to form a coherent or unique new whole. You can reorganize elements into a new pattern or structure through generating, planning, or producing.



2. Creative Thinking Skills

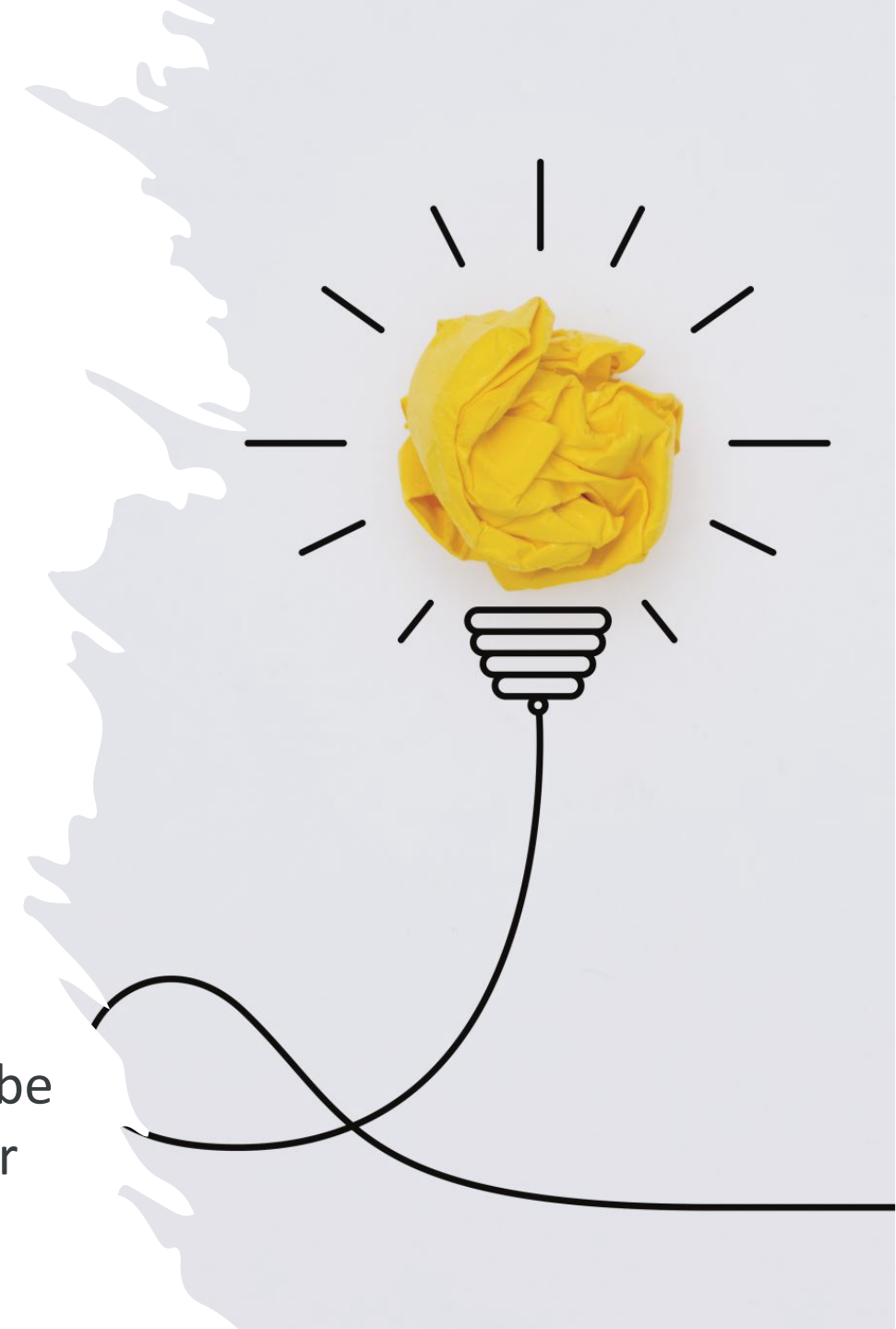
“Everybody has a creative potential and from the moment you can express this creative potential, **you can start changing the world.”**

Paulo Coelho, author and lyricist

Creative Thinking

- **Everyone has creative abilities.** It's true of everyone who fully expresses creative abilities as well as those who express them very little or not at all.
- **All humans are innately creative**, especially if creativity is understood as a problem-solving skill.
- **Creativity is inspired when there is a problem to solve.** As a creative thinker, you are curious, optimistic, and imaginative.

Creativity is inspired when there is a problem to solve. Considered as an act of problem-solving, creativity can be understood as a ***skill***—as opposed to an inborn talent or natural “gift”—that ***can be taught as well as learned.***



ACTIVITY: ASSESS YOUR CREATIVE PROBLEM-SOLVING SKILLS

Objective

- Evaluate your attitude toward problem-solving in the context of cultivating creative thinking.

Directions:

- Access *Psychology Today's* [Creative Problem-Solving Test](#) at the *Psychology Today* Web site.
- Read the introductory text, which explains how creativity is linked to fundamental qualities of thinking, such as flexibility and tolerance of ambiguity.
- Then advance to the questions by clicking on the “Take The Test” button. The test has 20 questions and will take roughly 10 minutes.
- After finishing the test, you will receive a Snapshot Report with an introduction, a graph, and a personalized interpretation for one of your test scores.
- Complete any further steps by following your instructor's directions.

Creative Thinking in Education



College is great ground for enhancing creative thinking skills.

These are some college activities that can stimulate creative thinking. Are any familiar to you?

- Design sample exam questions to test your knowledge as you study for a final.
- Devise a social media strategy for a club on campus.
- Propose an education plan for a major you are designing for yourself.
- Prepare a speech that you will give in a debate in your course.
- Develop a pattern for a costume in a theatrical production.
- Arrange audience seats in your classroom to maximize attention during your presentation.
- Arrange an eye-catching holiday display in your dormitory or apartment building.

Creative Thinking in Education



College is great ground for enhancing creative thinking skills. These are some college activities that can stimulate creative thinking. Are any familiar to you?

- Participate in a brainstorming session with your fellow musicians on how you will collaborate to write a musical composition.
- Draft a script for a video production that will be shown to several college administrators.
- Compose a set of requests and recommendations for a campus office to improve its customer service.
- Develop a marketing pitch for a mock business you are developing.
- Develop a comprehensive energy-reduction plan for your cohousing arrangement.

How to Stimulate Creative Thinking

- 1. Sleep on it.** Over the years, researchers have found that the REM sleep cycle boosts our creativity and problem-solving abilities, providing us with innovative ideas or answers to vexing dilemmas when we awaken. **Keep a pen and paper by the bed** so you can write down your nocturnal insights if they wake you up.
- 2. Go for a run or hit the gym.** Studies indicate that exercise stimulates creative thinking, and the brainpower boost lasts for a few hours.
- 3. Allow your mind to wander a few times every day.** Far from being a waste of time, daydreaming has been found to be an essential part of generating new ideas. If you're stuck on a problem or creatively blocked, think about something else for a while.



How to Stimulate Creative Thinking

- 4. Keep learning.** Studying something far removed from your area of expertise is especially effective in helping you think in new ways.
- 5. Put yourself in nerve-racking situations** once in a while to fire up your brain. Fear and frustration can trigger innovative thinking.
- 6. Keep a notebook with you** so you always have a way to record fleeting thoughts. They're sometimes the best ideas of all.





A Brainstorm of Tips for Creative Thinking

The best way to have a good idea is to have lots of ideas.

—Linus Pauling, double Nobel Laureate, chemist, biochemist, and peace campaigner





A Brainstorm of Tips for Creative Thinking

SENSING

- **Use all your senses** - see, taste, smell, touch, hear, think, speak.
- **Be a good observer** of people, nature, and events around you.

THINKING

- **Engage thinking** on the right side of your brain (intuition, open-mindedness, visual perception, rhythm . . .).
- **Change your interpretation** of an event, situation, behavior, person, or object.
- **Allow ideas to incubate.**
- **Be open** to insight as ideas pop into your mind.





A Brainstorm of Tips for Creative Thinking

IMAGINING

- Brainstorm by generating ideas with a group of people.
- Ask, “What would happen if . . .”
- Ask, “In how many different ways . . .”
- Develop ideas and expand their possibilities.
- Envision the future.





A Brainstorm of Tips for Creative Thinking

SPEAKING AND WRITING

- Use your words and your “voice” when conveying your original ideas.
- Avoid using clichés or overly familiar responses to questions or problems.
- Explain how your ideas move beyond the status quo and contribute to a discussion.
- Take notes.





A Brainstorm of Tips for Creative Thinking

DRAWING

- Use mind-mapping to capture ideas; start with a key concept and write it in the center of your page; use connecting lines, radiating from the central concept, and write down any connected or related ideas that come to you.
- Create pictures or drawings of situations (“rich pictures”) to show them in a different way.





A Brainstorm of Tips for Creative Thinking

LEARNING

Find ways to demonstrate your personal investment in projects.

Gather knowledge and conduct research.

Have more fun learning!

MOVING

Do physical activities to engage the creative areas of your brain and think differently.

RESTING

Take breaks.



The background is a collage of grey puzzle pieces. Several pieces feature a black question mark. In the center, a puzzle piece contains a blue line drawing of a lit lightbulb with radiating lines. The text "Creative Thinking" is centered over the lightbulb piece.

Creative Thinking

Fiction and Facts

Creative Thinking - Fiction and Facts

FICTION

- Every problem has only one solution (or one right answer).

FACTS

- Most problems can be solved in any number of ways.
- If you discover a solution that works, it's a good solution.
- Other people may think up solutions that differ from yours, but that doesn't make your solution wrong or unimportant.

Creative Thinking - Fiction and Facts

FICTION

- The best answer or solution or method has already been discovered.

FACTS

- Look at the history of any solution and you'll see that improvements, new solutions, and new right answers are always being found.
- The ox or horse, the cart, the wagon, the train, the car, the airplane, the jet, the space shuttle? What is the best and last?

Creative Thinking - Fiction and Facts

FICTION

- Creative answers are technologically complex.

FACTS

- Only a few problems require complex technological solutions.
- Most problems you'll encounter need only a thoughtful solution involving personal action and perhaps a few simple tools.
- Even many problems that seem to require technology can be addressed in other ways.

Creative Thinking - Fiction and Facts

FICTION

- Ideas either come or they don't. Nothing will help - certainly not structure.

FACTS

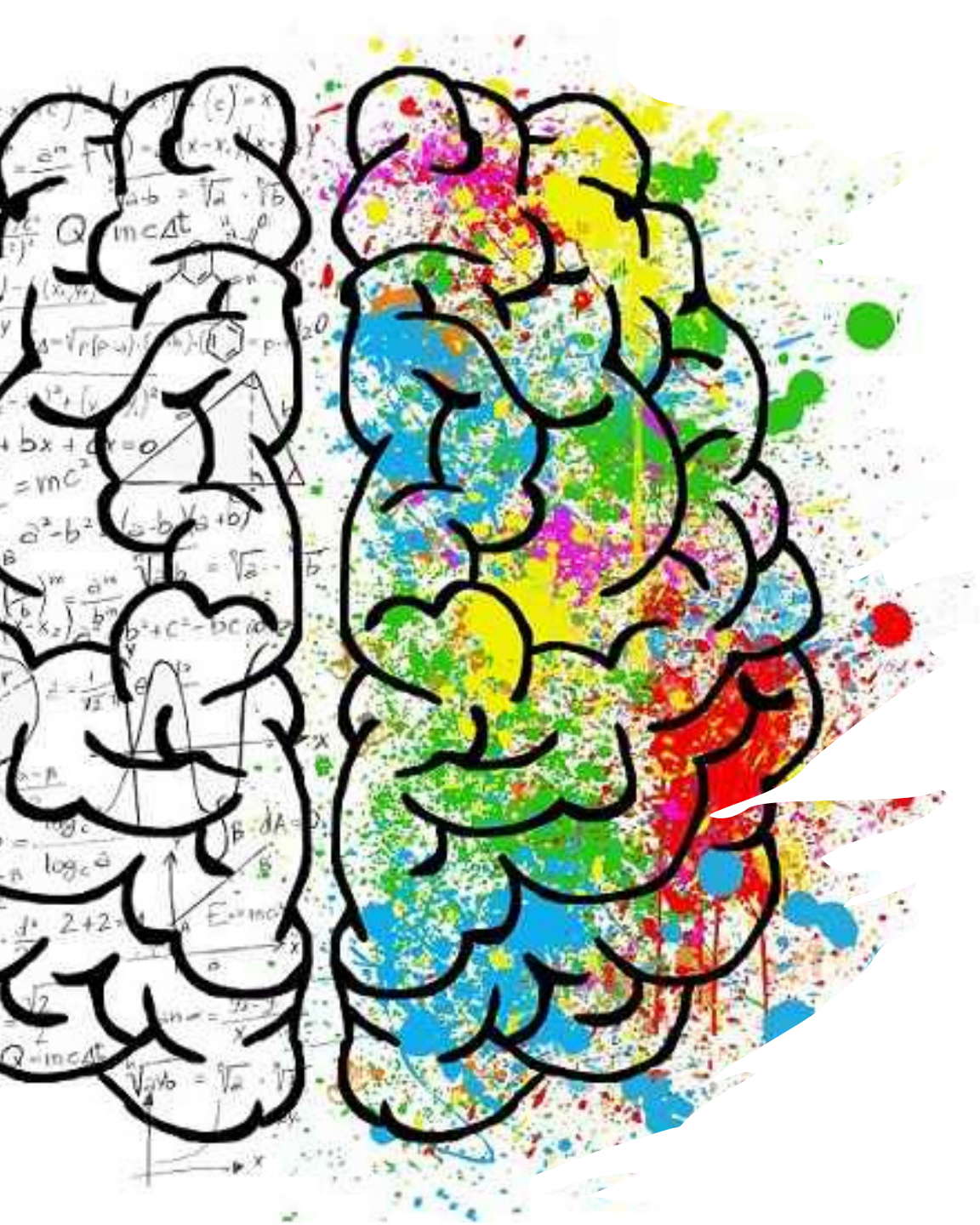
- There are many successful techniques for generating ideas. One important technique is to include structure.
- Create guidelines, limiting parameters, and concrete goals for yourself that stimulate and shape your creativity.



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THINKING AND ANALYSIS (cont.)



Chapter outlines

THINKING AND ANALYSIS

2. Creative Thinking Skills (cont.)



LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Describe the role of creative thinking skills in problem-solving
- Identify technology tools that enhance our learning
- Explain how technology skills relate to critical/creative thinking skills
- Examine online learning in the context of organizing, communicating, reading, and researching online
- Assess our readiness to use technology



Problem-Solving with Creative Thinking

- Creative problem-solving is a type of problem-solving. It involves searching for new and novel solutions to problems.
- Unlike critical thinking, which scrutinizes assumptions and uses reasoning, creative thinking is about generating alternative ideas - practices and solutions that are unique and effective.
- It's about facing sometimes muddy and unclear problems and seeing how “things” can be done differently - how new solutions can be imagined.



Project start up "Social initiative":

Discuss in small groups, propose a creative social initiative, to solve any problem our society.

Rules of Brainstorming



Defer Judgment



Encourage Wild Ideas



Build on the Ideas of Others



Stay Focused on the Topic



One Conversation at a Time



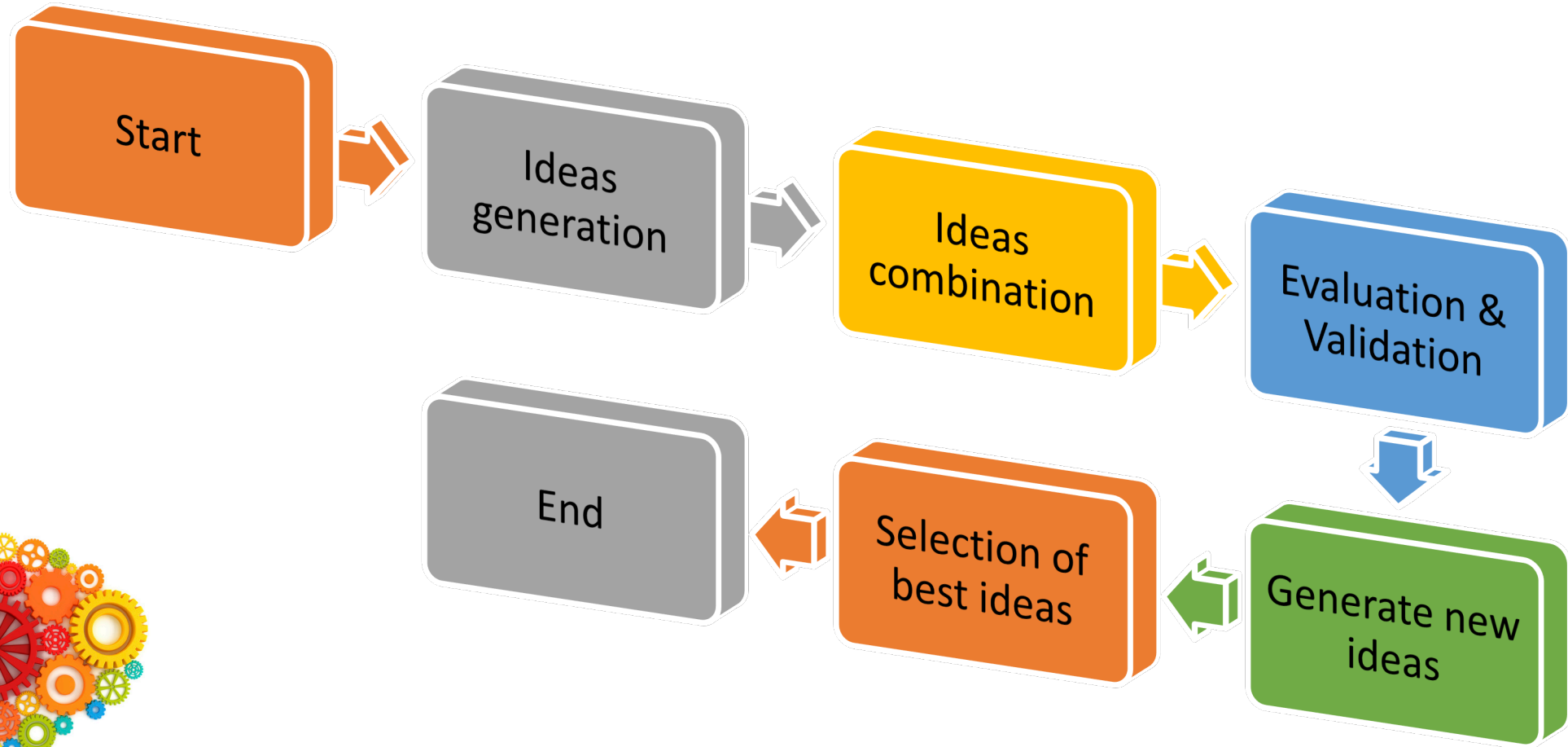
Be Visual



Go for Quantity



Brainstorming technique



it's
Q&A
TIME!



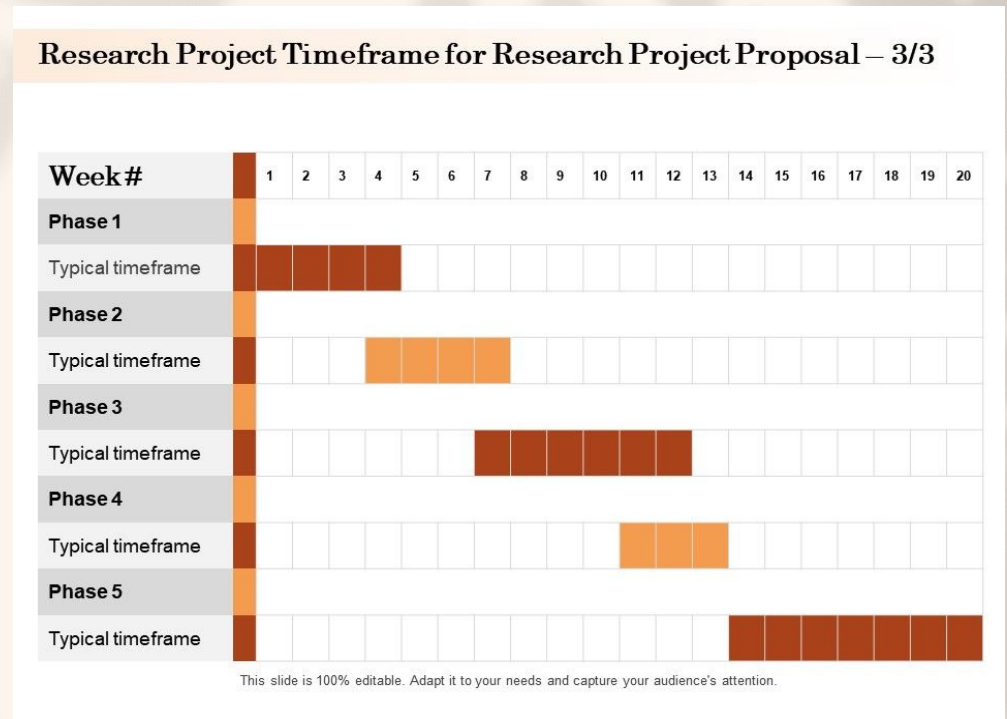
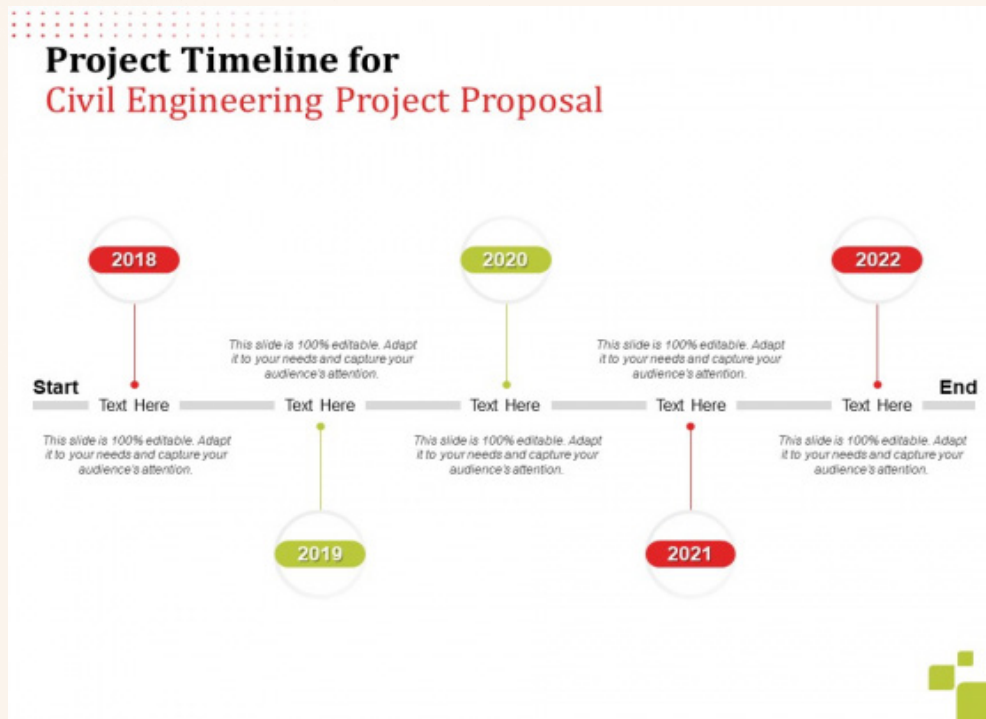


BUSINESS PROPOSAL (cont.)

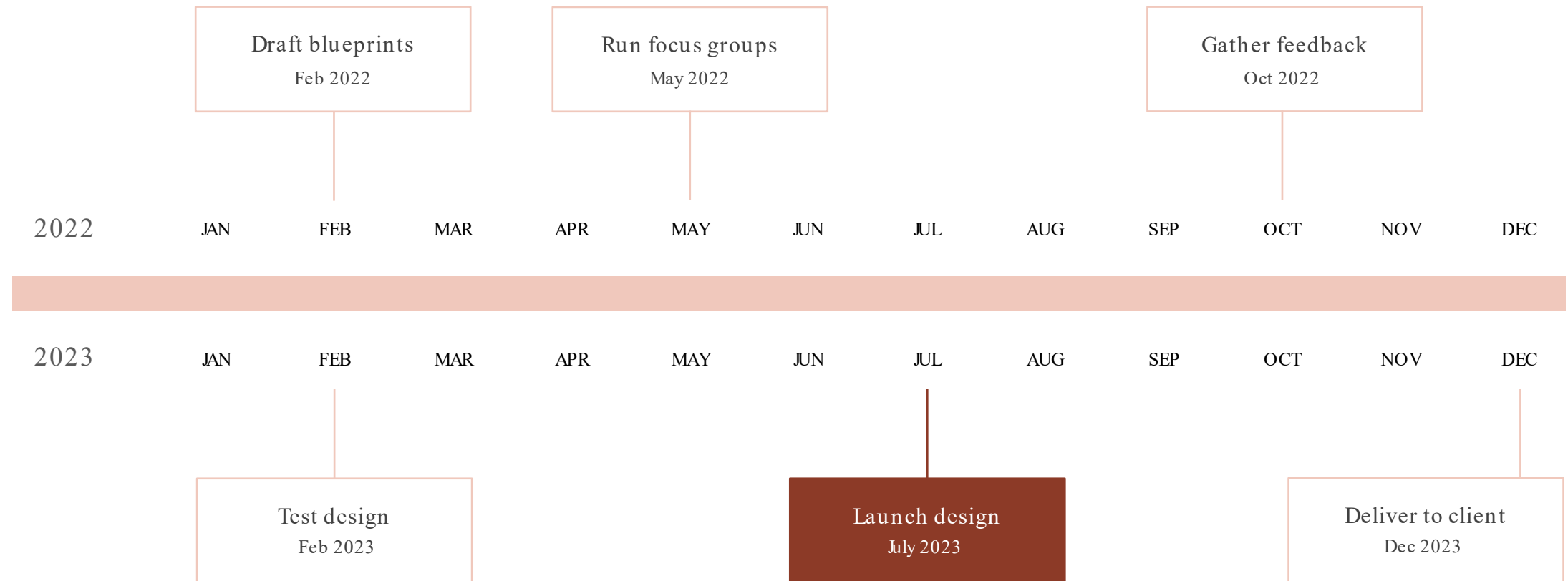
TRADITIONAL CATEGORIES OF PROPOSAL

7. TIMELINE

A clear presentation, often with visual aids, of the process, from start to finish, with specific, dated benchmarks noted.



Sample of a Milestone Timeline



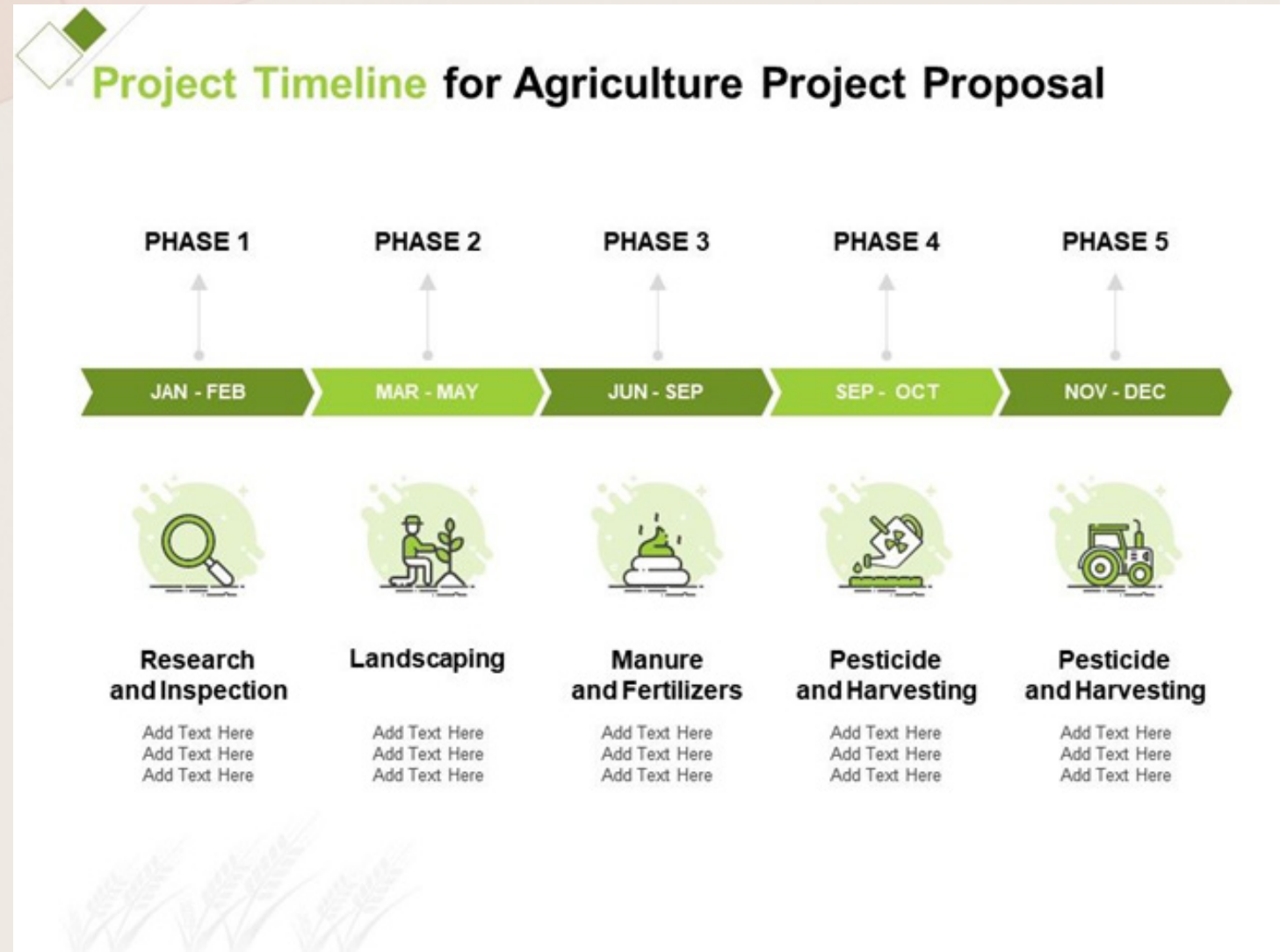
8. The Marketing Plan

- [illegible]



Let's practice!

Let's discuss and write the timeline and marketing plan of the project.



TRADITIONAL CATEGORIES OF PROPOSAL

9. FINANCE



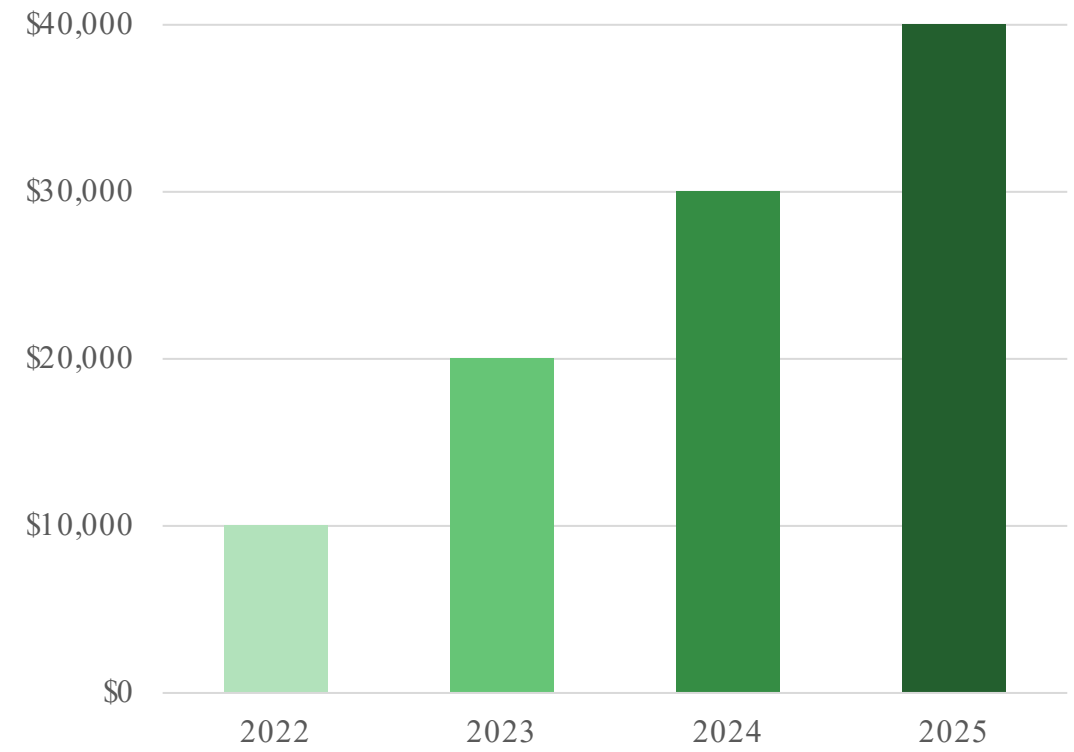
- What are the initial costs, when can revenue be anticipated, when will there be a return on investment (if applicable)?
- Again, the proposal may involve a one-time fixed cost, but if the product or service is to be delivered more than once and extended financial plan noting costs across time is required.

Sample Financials

Key metrics

	Clients	Orders	Gross revenue	Net revenue
2022	10	1100	\$10,000	\$7,000
2023	20	200	\$20,000	\$16,000
2024	30	300	\$30,000	\$25,000
2025	40	400	\$40,000	\$30,000

Revenue by year



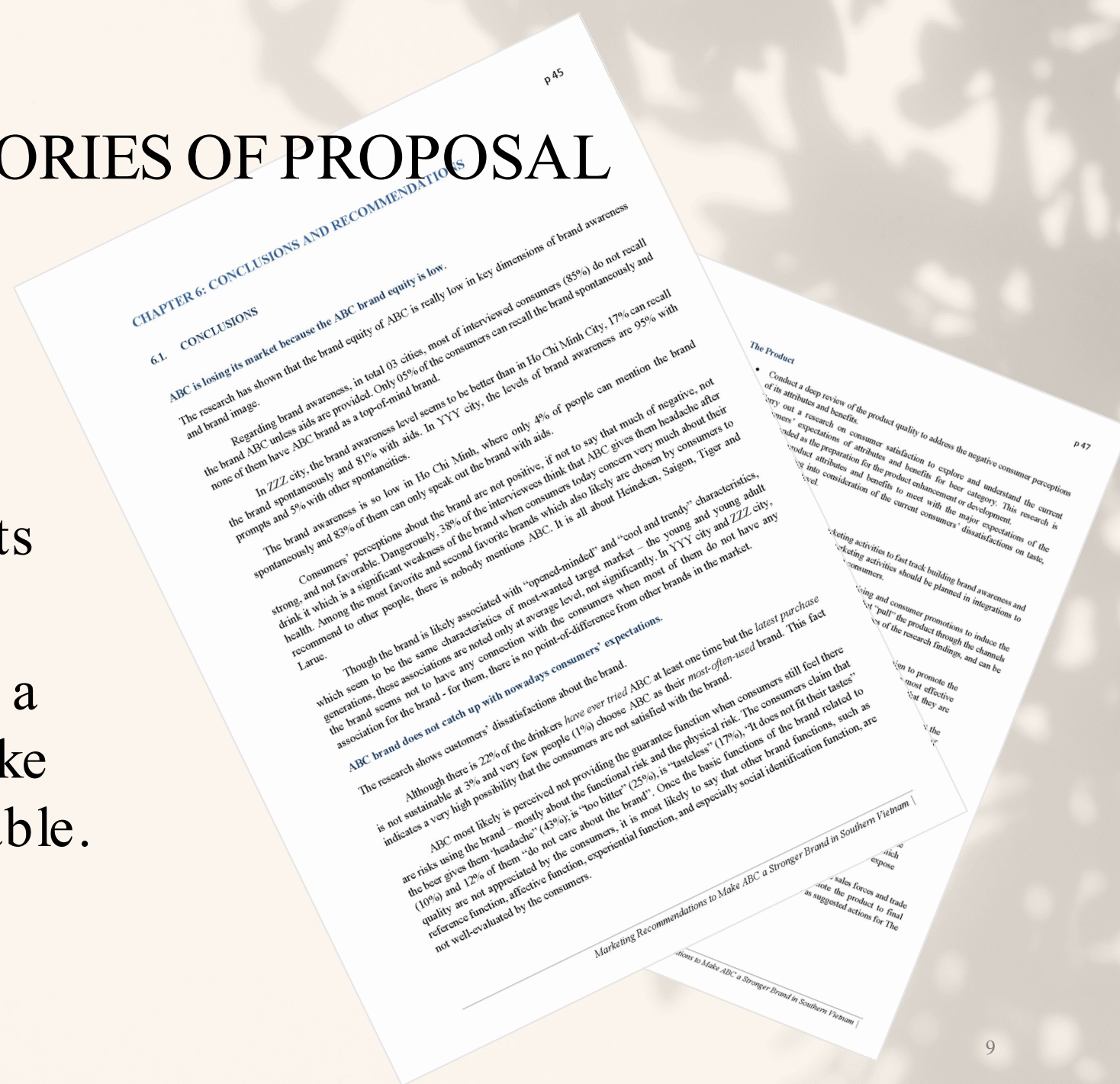
Sample Financials table

DESCRIPTION	YEAR 1	YEAR 2	YEAR 3	DELTA
Income				
Users	50,000	400,000	1,600,000	
Sales	500,000	4,000,000	16,000,000	
Average price per sale	75	80	90	
Revenue @15%	5,625,000	48,000,000	216,000,000	
Gross profit	5,625,000	48,000,000	216,000,000	
Expenses				
Sales &marketing	5,062,500	38,400,000	151,200,000	70%
Customer service	1,687,500	9,600,000	21,600,000	10%
Product development	562,500	2,400,000	10,800,000	5%
Research	281,250	2,400,000	4,320,000	2%
Total expenses	7,593,750	52,800,000	187,920,000	

TRADITIONAL CATEGORIES OF PROPOSAL

10. CONCLUSION

- Like a speech or essay, restate your main points clearly.
- Tie them together with a common theme and make your proposal memorable.



Let's practice!

Let's discuss and compose the finance and conclusion of the project.

CHAPTER 6: CONCLUSIONS AND RECOMMENDATIONS

6.1. CONCLUSIONS

ABC is losing its market because the ABC brand equity is low. The research has shown that the brand equity of ABC is really low in key dimensions of brand awareness and brand image.

Regarding brand awareness, in total 03 cities, most of interviewed consumers (85%) do not recall the brand ABC unless aids are provided. Only 05% of the consumers can recall the brand spontaneously and none of them have ABC brand as a top-of-mind brand.

In ZZZ city, the brand awareness level seems to be better than in Ho Chi Minh City. 17% can recall the brand spontaneously and 81% with aids. In YYY city, the levels of brand awareness are 95% with prompts and 5% with other spontaneous.

The brand awareness is so low in Ho Chi Minh, where only 4% of people can mention the brand spontaneously and 85% of them can only speak out the brand with aids.

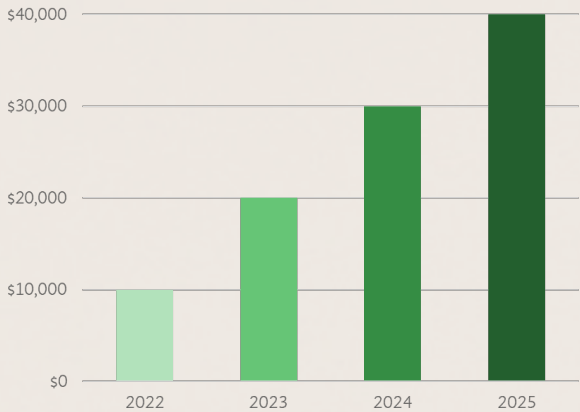
Consumers' perceptions about the brand are not positive, if not to say that much of negative, not strong and not favorable. Dangerously, 38% of the interviewees think that ABC gives them headache after drink it which is a significant weakness of the brand when consumers today concern very much about their health. Among the most favorite and second favorite brands which also likely are chosen by consumers to recommend to other people, there is nobody mentions ABC. It is all about Heineken, Saigon, Tiger and Larue.

Though the brand is likely associated with "opened-minded" and "cool and trendy" characteristics, these associations are noted only at average level, not significantly. In YYY city and ZZZ city, consumers do not have any connection with the consumers when most of them do not have any connection with the brand. There is no point-of-difference from other brands in the market.

Key metrics

	Clients	Orders	Gross revenue	Net revenue
2022	10	1100	\$10,000	\$7,000
2023	20	200	\$20,000	\$16,000
2024	30	300	\$30,000	\$25,000
2025	40	400	\$40,000	\$30,000

Revenue by year



DESCRIPTION	YEAR1	YEAR2	YEAR3	DELTA
Income				
Users	50,000	400,000	1,600,000	
Sales	500,000	4,000,000	16,000,000	
Average price per sale	75	80	90	
Revenue @ 15%	5,625,000	48,000,000	216,000,000	
Gross profit	5,625,000	48,000,000	216,000,000	
Expenses				
Sales & marketing	5,062,500	38,400,000	151,200,000	70%
Customer service	1,687,500	9,600,000	21,600,000	10%
Product development	562,500	2,400,000	10,800,000	5%
Research	281,250	2,400,000	4,320,000	2%
Total expenses	7,593,750	52,800,000	187,920,000	

❖ Ethos, Pathos, and Logos

- **Ethos** refers to credibility
- **Pathos** refers passion and enthusiasm
- **Logos** refers logic or reason



Let's practice! Create your own business logo.

Activity

Create your Team Project's Logo



❖ Professional

A professional document is a base requirement:

- Should be no errors in spelling or grammar
- Be concise, accurate, and clearly referenced
- Should be easy to find and clearly relevant, including contact information.



Persuasive proposals

Are often brief, even limited to one page:

The one-page proposal has been one of the keys to my business success, and it can be invaluable to you too. Few decision-makers can ever afford to read more than one page when deciding if they are interested in a deal or not. This is even more true for people of a different culture or language.

Adnan Khashoggi, a successful multibillionaire (Riley, 2002)

Two Types of Business Proposals

Solicited

- If you have been asked to submit a proposal it is considered solicited.
- May come in the form of a direct verbal or written request

Unsolicited

- Unsolicited proposals are the “cold calls” of business writing.
- Required a thorough understanding of the market, product and/or service, and their presentation is typically general rather than customer-specific.



Sample Business Proposal

The Writing Help Tools Center is a commercial enterprise, and offers a clear (and free) example of a business proposal here:

<http://www.writinghelp-central.com/sample-business-proposal.html>

Finalize your project proposal

- Prepare a draft business proposal in no more than 2 pages.
- Do not include actual contact information. Just as the example has employees named after colors, your (imaginary) company should have contact information that does not directly link to real businesses or you as an individual.

Q&A





BUSINESS PROPOSAL



BUSINESS PROPOSAL

Marketing Biz



DD SIMPLE

1387 Chandler Drive,
Hermitage
Missouri

Prepared By,

Tobias Ethan,
Marketing Manager
Marketing Biz

- Describe the basic elements of a business proposal.
 - Discuss the main goals of a business proposal.
- Identify effective strategies to use in a business proposal.

LEARNING OBJECTIVES



INTRODUCTION

- In order to be successful in business and industry, you should be familiar with the business proposal.
- Business proposals are documents designed to make a persuasive appeal to the audience to achieve a defined outcome, often proposing a solution to a problem.

BUSINESS PROPOSAL



COMMON PROPOSAL ELEMENTS

1. The Ideas
2. Traditional Categories
3. Ethos, Pathos, and Logos
4. Professional

**BUSINESS
PROPOSAL**



COMMON PROPOSAL ELEMENTS

THE IDEA

Effective business proposals are built around a great idea or solution.

- What makes your idea different or unique?
- How can you better meet the needs of the company that other vendors?
- What makes you so special?



COMMON PROPOSAL ELEMENTS

TRADITIONAL CATEGORIES OF PROPOSAL

1. Cover Page
2. Executive Summary
3. Background
4. Proposal
5. Market Analysis
6. Benefits
7. Timeline
8. Marketing Plan
9. Finance
10. Conclusion



COMMON PROPOSAL ELEMENTS

TRADITIONAL CATEGORIES OF PROPOSAL

Cover Page	Title page with name, title, date, and specific reference to request for proposal if applicable.
Executive Summary	Like an abstract in a report, this is a one- or two-paragraph summary of the product or service and how it meets the requirements and exceeds expectations.
Background	Discuss the history of your product, service, and/or company and consider focusing on the relationship between you and the potential buyer and/or similar companies.
Proposal	The idea. Who, what, where, when, why, and how. Make it clear and concise. Don't waste words, and don't exaggerate. Use clear, well-supported reasoning to demonstrate your product or service.
Market Analysis	What currently exists in the marketplace, including competing products or services, and how does your solution compare?



COMMON PROPOSAL ELEMENTS

TRADITIONAL CATEGORIES OF PROPOSAL

Benefits	How will the potential buyer benefit from the product or service? Be clear, concise, specific, and provide a comprehensive list of immediate, short, and long-term benefits to the company.
Timeline	A clear presentation, often with visual aids, of the process, from start to finish, with specific, dated benchmarks noted.
Marketing Plan	Delivery is often the greatest challenge for Web-based services—how will people learn about you? If you are bidding on a gross lot of food service supplies, this may not apply to you, but if an audience is required for success, you will need a marketing plan.
Finance	What are the initial costs, when can revenue be anticipated, when will there be a return on investment (if applicable)? Again, the proposal may involve a one-time fixed cost, but if the product or service is to be delivered more than once and extended financial plan noting costs across time is required.
Conclusion	Like a speech or essay, restate your main points clearly. Tie them together with a common theme and make your proposal memorable.

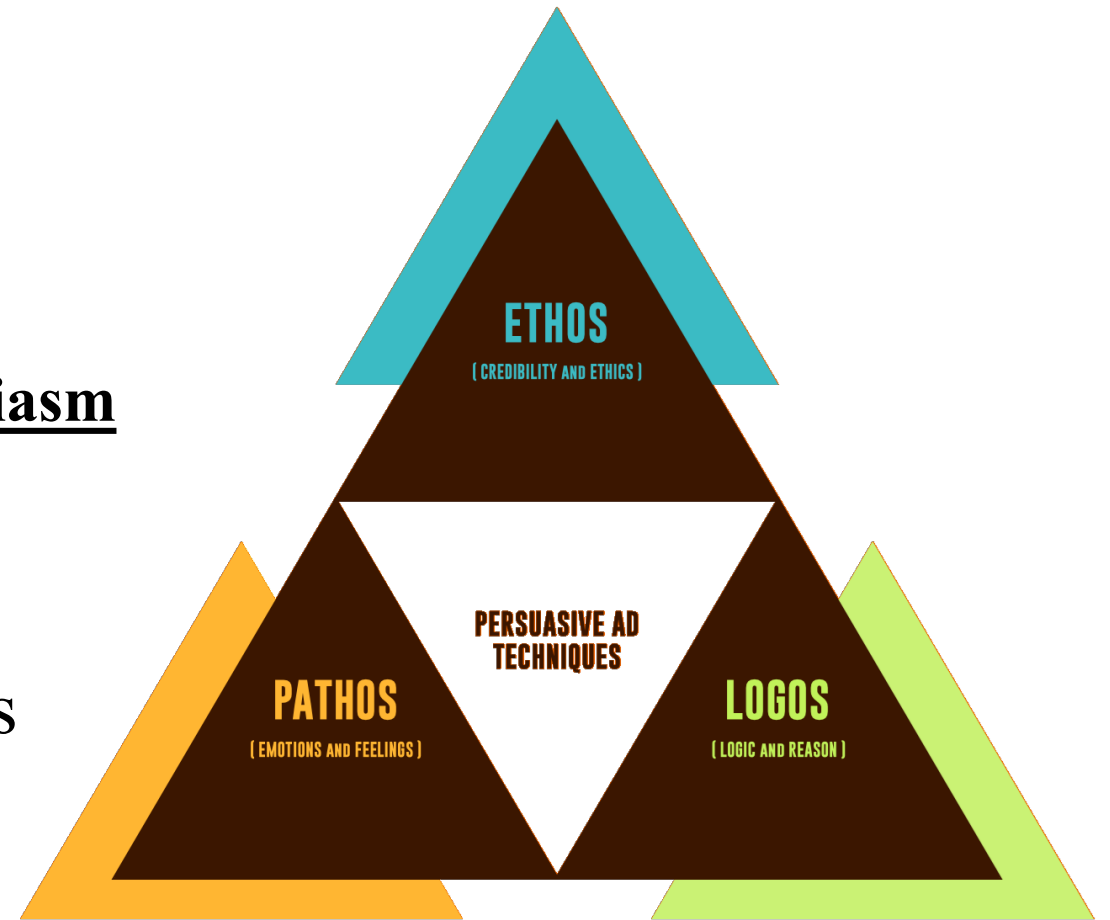


COMMON PROPOSAL ELEMENTS

❖ Ethos, Pathos, and Logos

- **Ethos** refers to credibility
- **Pathos** prefer to passion and enthusiasm
- **Logos** prefer to logic or reason

All three elements are integral parts of your business proposal that require your attention.



COMMON PROPOSAL ELEMENTS

❖ Professional

- A professional document is a base requirement. If it is less than professional, you can count on its prompt dismissal.
- There should be no errors in spelling or grammar, and all information should be concise, accurate, and clearly referenced when appropriate.



PRACTICE:

CHOOSE IDEA FOR "SOCIAL INITIATIVE" PROJECT

Decide on an idea your team will work on, make sure it's a creative, groundbreaking idea.

THE-BEST-IDEA!



TRADITIONAL CATEGORIES OF PROPOSAL

1. COVER PAGE

Title page with name, title, date, and specific reference to request for proposal if applicable.

[Type the company name here]

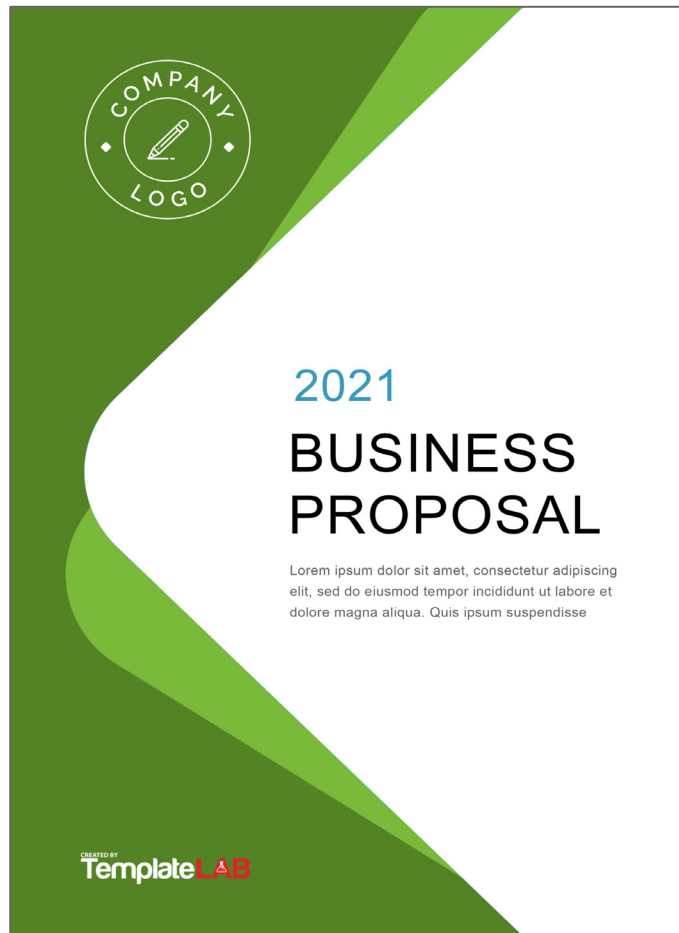
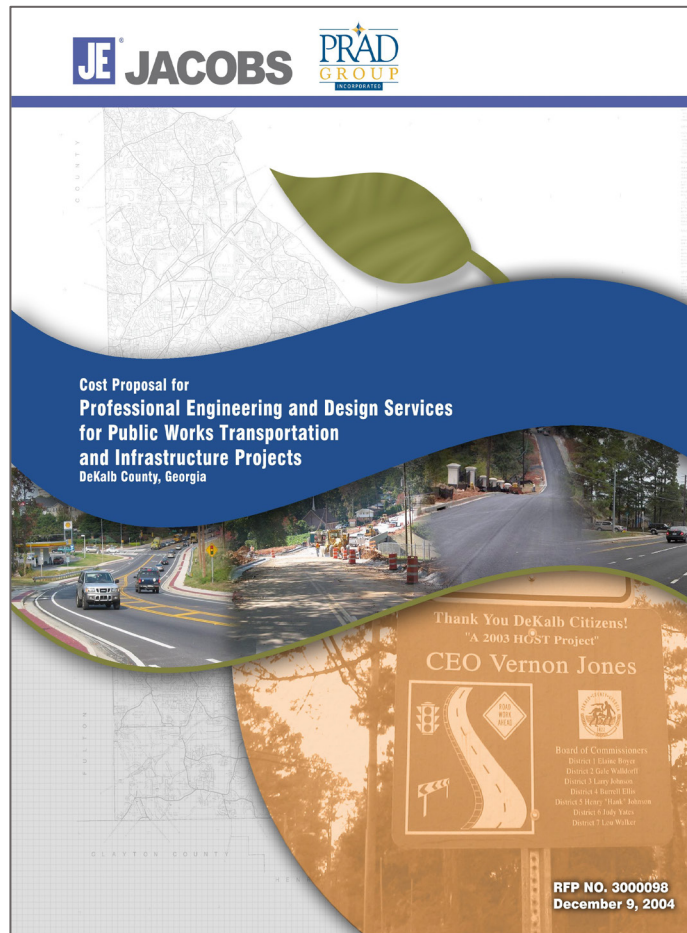
[Type the document title]
[Type the document subtitle]

John Miller
System Analyst

[Write a description or summary of the documents enclosed. This is not the preface or forewords section, this description is an overview of the content enclosed]



EXAMPLES OF COVER PAGE



TRADITIONAL CATEGORIES OF PROPOSAL

2. EXECUTIVE SUMMARY

Like an abstract in a report, this is a one- or two-paragraph summary of the product or service and how it meets the requirements and exceeds expectations.

Abstract

Through critical and deep study, this report examines the types of production systems that are used in apparel industry of Pakistan. It shows the analysis and comparison of production systems with respect to cost, quality and productivity. Since the function of sewing department mainly depends on the type of production system, its selection needs to be done carefully.

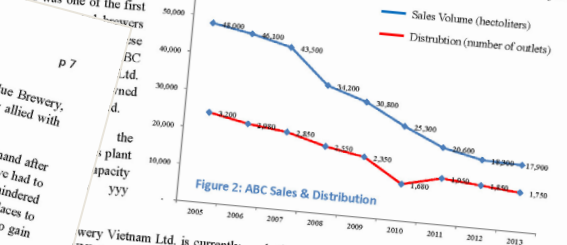
This study is based on the analysis of the handling time required to complete the operations of a polo shirt, as the needle time is constant in all types of systems. This thesis also focuses on the positive and negative aspects of production systems. It demonstrates that the industry has made significant progress towards comprehending the necessity of proper production system which has implications on the overall productivity of an apparel mill.



TRADITIONAL CATEGORIES OF PROPOSAL

3. BACKGROUND

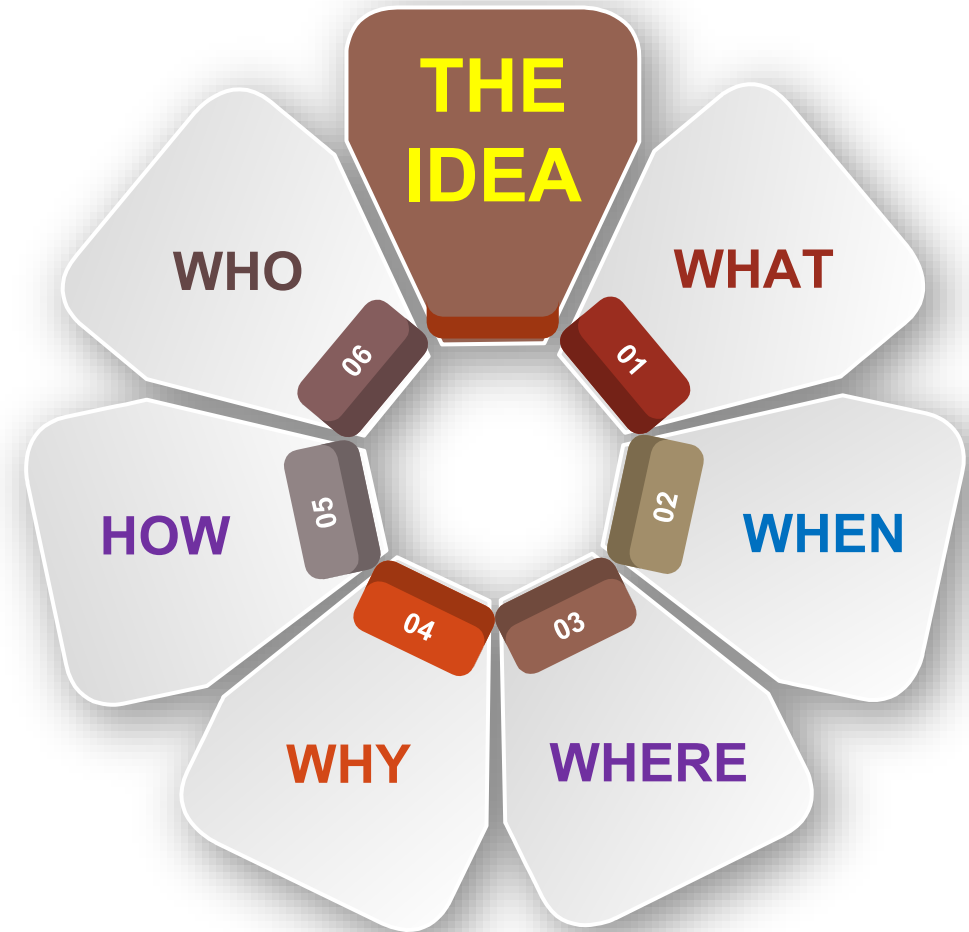
Discuss the history of your product, service, and/or company and consider focusing on the relationship between you and the potential buyer and/or similar companies.



TRADITIONAL CATEGORIES OF PROPOSAL

4. PROPOSAL

- Make it clear and concise.
- Don't waste words,
- Don't exaggerate.
- Use clear, well-supported reasoning to demonstrate your product or service.



TRADITIONAL CATEGORIES OF PROPOSAL

5. MARKET ANALYSIS

What currently exists in the marketplace, including competing products or services, and how does your solution compare?



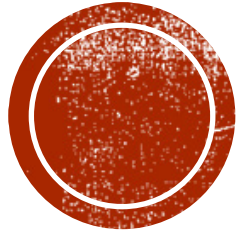
TRADITIONAL CATEGORIES OF PROPOSAL

6. BENEFITS

How will the potential buyer benefit from the product or service?

- Clear, Concise, Specific, Short
- Provide a comprehensive list of immediate and long-term benefits





LET'S START PRACTICING

Small group (5-7 members): Write a proposal from an idea of "Social initiative" project

Q&A

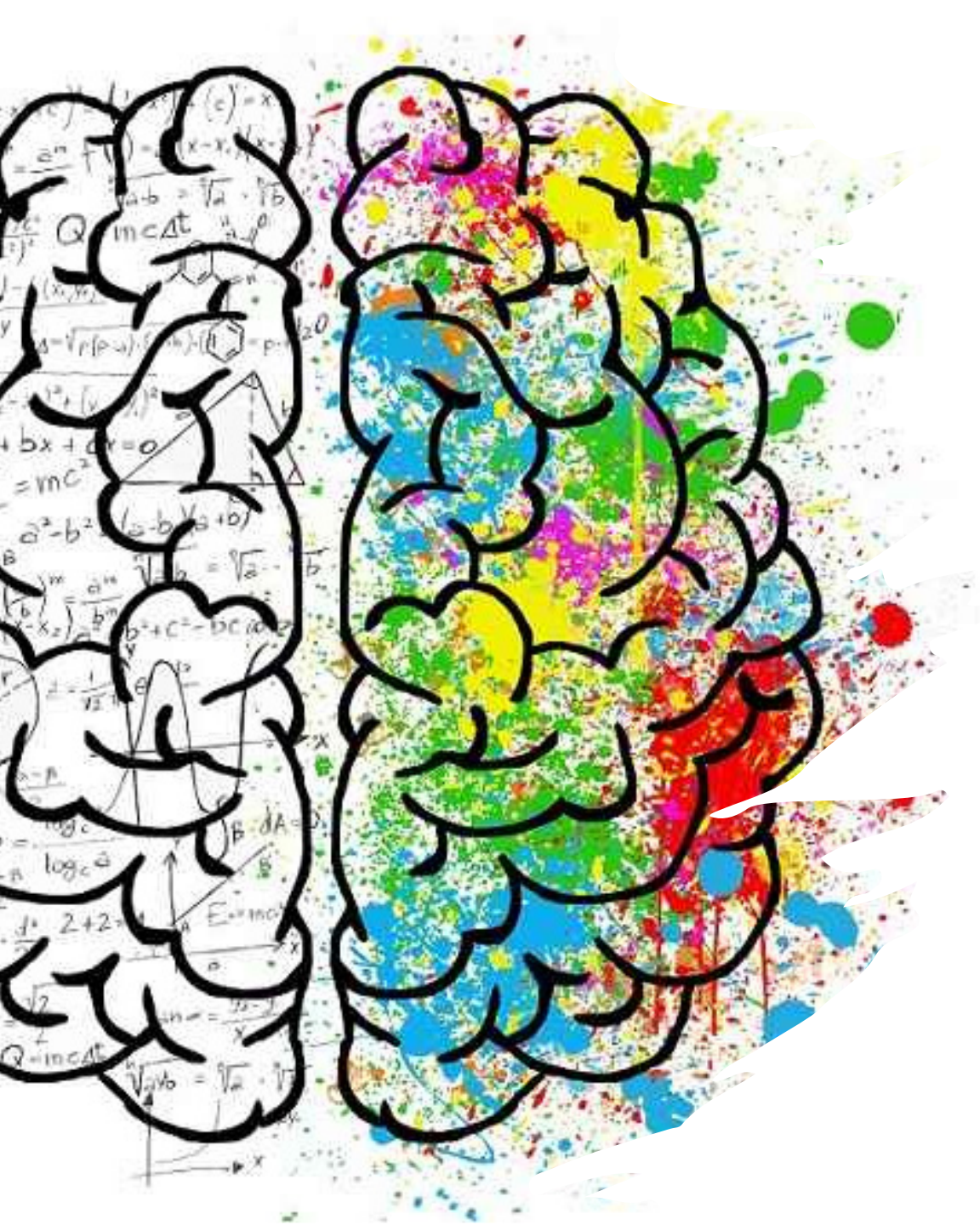




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THINKING AND ANALYSIS (cont.)



Chapter outlines

THINKING AND ANALYSIS

4. Critical Thinking Skills

A photograph of three people in a meeting. A woman with glasses and a floral blazer is smiling while looking at a laptop. A man in a blue shirt is leaning over her, and another man with glasses is in the foreground, also looking at the laptop. The scene is set in a modern office with a laptop and papers on the desk.

Critical Thinking Skills

“The essence of the independent mind lies not in what it thinks, but in how it thinks.”

Christopher Hitchens, author and journalist



LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Define critical thinking
- Describe the role that logic plays in critical thinking
- Describe how critical thinking skills can be used to problem-solve
- Describe how critical thinking skills can be used to evaluate information
- Identify strategies for developing yourself as a critical thinker.

What Is Critical Thinking?

- Critical thinking is clear, reasonable, reflective thinking focused on deciding what to believe or do.
- It involves being **skeptical and challenging assumptions**, rather than simply memorizing facts or blindly accepting what you hear or read.
- Asking probing questions like:

“How do we know?”

“Is this true in every case or just in this instance?”



Critical Thinking IS

Skepticism

Examining assumptions

Challenging reasoning

Uncovering biases



Critical Thinking is NOT

Memorizing

Group thinking

Blind acceptance of authority

ACTIVITY: SELF-ASSESS YOUR CRITICAL THINKING STRATEGIES

Objectives

- Assess your basic understanding of the skills involved in critical thinking.

Directions

- Visit the [Quia Critical Thinking Quiz](#) page and click on Start Now (you don't need to enter your name). Select the best answer for each question, and then click on Submit Answers. A score of 70 percent or better on this quiz is considering passing.
- Based on the content of the questions, do you feel you use good critical thinking strategies in college? In what ways might you improve as a critical thinker?





Critical Thinking and Logic

Critical thinking is fundamentally a process of questioning information and data.

You may question the information you read in a textbook, or you may question what a politician or a professor or a classmate says. You can also question a commonly-held belief or a new idea.

With critical thinking, anything and everything is subject to question and examination for the purpose of logically constructing reasoned perspectives.



What Is Logic, and Why Is It Important in Critical Thinking?

The word **logic** comes from the Ancient Greek “**logike**”, referring to **the science or art of reasoning**.

Using logic, a person **evaluates arguments and reasoning** and strives to **distinguish between good and bad reasoning**, or **between truth and falsehood**.

Using logic, you can **evaluate ideas or claims** people make, **make good decisions**, and **form sound beliefs** about the world.



Questions of Logic in Critical Thinking

- 1. What's happening?** Gather the basic information and begin to think of questions.
- 2. Why is it important?** Ask yourself why it's significant and whether or not you agree.
- 3. What don't I see?** Is there anything important missing?
- 4. How do I know?** Ask yourself where the information came from and how it was constructed.
- 5. Who is saying it?** What's the position of the speaker and what is influencing them?
- 6. What else? What if?** What other ideas exist and are there other possibilities?

Questions a Critical Thinker Asks

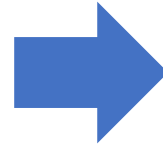




Problem-Solving with Critical Thinking



Critical thinking and
problem-solving go
hand-in-hand



Refer to using knowledge,
facts, and data to solve
problems effectively.

Examples of using critical thinking to problem-solve

- Your classmate was upset and said some unkind words to you, which put a crimp in the relationship. You try to see through the angry behaviors to determine how you might best support the roommate and help bring the relationship back to a comfortable spot.
- Your campus club has been languishing on account of lack of participation and funds. The new club president, though, is a marketing major and has identified some strategies to interest students in joining and supporting the club. Implementation is forthcoming.
- Your math teacher sees that the class is not quite grasping a concept. She uses clever questioning to dispel anxiety and guide you to new understanding of the concept.



Problem-Solving Action Checklist

STRATEGIES

ACTION CHECKLIST

1. Define the problem

- Identify the problem
 - Provide as many supporting details as possible
 - Provide examples
 - Organize the information logically
-

2. Identify available solutions

- Use logic to identify your most important goals
 - Identify implications and consequences
 - Identify facts
 - Compare and contrast possible solutions
-

3. Select your solution

- Use gathered facts and relevant evidence
 - Support and defend solutions considered valid
 - Defend your solution
-

Evaluating Information with Critical Thinking

- Evaluating information can be one of the most complex tasks.
- But if you utilize the following four strategies, you will be well on your way to success:
 - A. Read for understanding by using text coding
 - B. Examine arguments
 - C. Clarify thinking
 - D. Cultivate “habits of mind”

A. Read for Understanding Using Text Coding

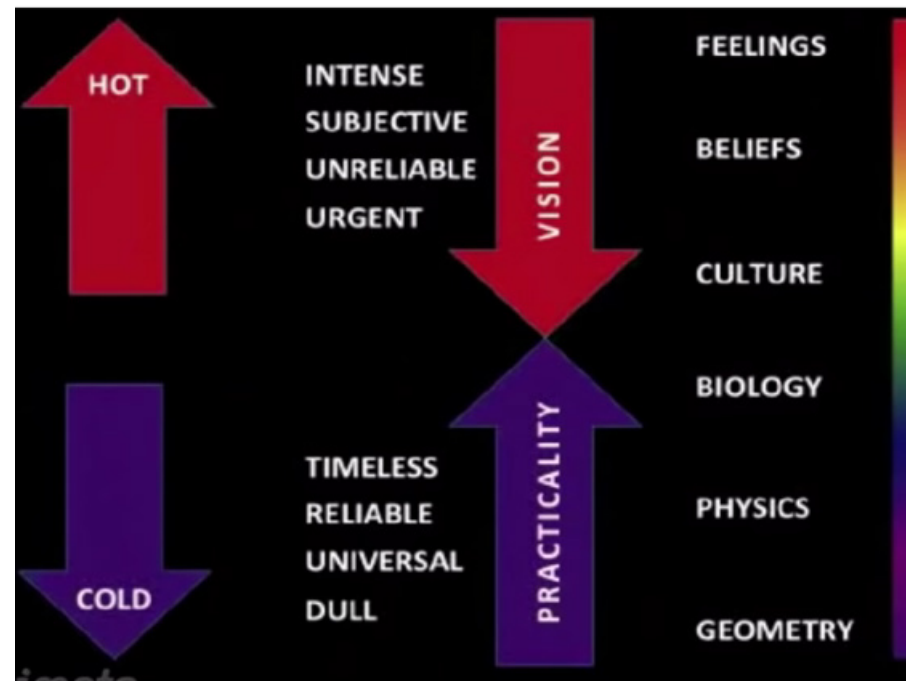
- Text coding is a way of tracking your thinking while reading
- With text coding, mark important arguments and key facts

See more text coding from [PBWorks](#) and [Collaborative for Teaching and Learning](#).

Shorthand	Meaning
!	Important
L	Learned something new
!	Big idea surfaced
*	Interesting or important fact
?	Dig deeper
✓	Agree
≠	Disagree

B. Examine Arguments

Use the spectrum of authority strategy



https://youtu.be/9G5xooMN2_c

C. Clarify Thinking

When you use critical thinking to evaluate information, you **need to clarify your thinking to yourself and likely to others.**

- What is the purpose?
- What question are we trying to answer?
- What point of view is being expressed?
- What assumptions are we or others making?

C. Clarify Thinking (cont.)

When you use critical thinking to evaluate information, you **need to clarify your thinking to yourself and likely to others.**

- What are the facts and data we know, and how do we know them?
- What are the concepts we're working with?
- What are the conclusions, and do they make sense?
- What are the implications?

D. Cultivate “Habits of Mind”



“Habits of mind” are the **personal commitments, values, and standards** you have about **the principle of good thinking.**

Do you approach problems with an open mind, a respect for truth, and an inquiring attitude?

Try to work these qualities into your daily life:

- Being receptive to having your opinions changed
- Having respect for others
- Being independent
- Not accepting something is true until you've had the time to examine the available evidence



Try to work these qualities into your daily life:

- Being fair-minded
- Having respect for a reason
- Having an inquiring mind
- Not making assumptions
- Questioning your own conclusions





Developing Yourself As a Critical Thinker

- Reflect and practice
 - Use wasted time
- Redefine the way you see things
- Analyze the influences on your thinking and in your life
 - Express yourself
- Enhance your wellness

Practice:

Evaluation of
project ideas
"Social
Initiatives"

- Use critical thinking skills to **challenge other groups' social initiatives ideas** (from activity 1 of Creative Thinking Skills)
- **Point out weaknesses and difficulties** in implementing ideas in life

